

Regime-Locked Measurement as a General Structural Constraint:

Cross-Domain Framework, Operational Meta-Regime Protocol,
and Quantitative Regime-Appropriateness Scoring

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Abstract. Every measurement system operates within an implicit *regime*: a bounded domain of validity defined by the instrument’s design and assumptions. Outside this domain, the instrument is not imprecise—it is structurally blind. This paper formalises *regime-locked measurement* as a general structural constraint, demonstrates a shared structural pattern across six independent domains (observational cosmology, AI alignment, cyber security, intelligence analysis, financial economics, and consciousness research), and provides three operational contributions. First, a quantitative *regime-appropriateness score* $\alpha(\phi, \mathcal{R}) \in [0, 1]$ that replaces binary classification with continuous evaluation. Second, a taxonomy of blind-set types (ontological, methodological, instrumental) with distinct detection strategies. Third, a five-step *meta-regime protocol* with a worked example in AI alignment that identifies convergence drift—a class of alignment change invisible to current deviation-focused instruments. We provide an explicit falsification criterion and apply the framework’s constraints to itself.

1 Introduction

1.1 The Problem

The history of science is a history of measurement. Each leap in understanding has been preceded by a leap in what could be *observed*. Yet implicit in every measurement is a constraint rarely made explicit: the instrument does not measure reality—it measures the projection of reality onto its operational domain.

This observation is not new. Heisenberg’s uncertainty principle formalises it for conjugate variables. Kuhn’s paradigm theory describes it for scientific communities. Bayesian model comparison (AIC, BIC, Bayes factors) addresses it for statistical inference. What has been missing is a *cross-domain formal framework* that treats instrument-defined observability as a general structural constraint—one that manifests whether the “instrument” is a telescope, a neural network evaluation suite, or a national intelligence apparatus.

This paper proposes such a framework, drawing on regime architecture concepts from thermodynamic cosmology [1, 2, 3], generalised to a domain-independent formulation.

1.2 Scope and Claims

We distinguish three levels of claim:

Table 1: Epistemic status of claims.

Level	Status	Content
Definition	Formal	Regime, regime-locked measurement, CDVC, meta-regime, α -score, blind-set taxonomy
Hypothesis	Testable	A shared structural pattern operates across domains; meta-regime evaluation can detect blind spots; α -scoring outperforms binary classification
Conjecture	Open	S_0/S_1 observational equivalence at gradient extremes

We explicitly avoid claiming universality. The framework is proposed as a *widely applicable structural constraint* whose scope must be determined empirically, domain by domain.

2 Formal Framework

2.1 Definition: Regime

A **regime** $\mathcal{R}$ is a bounded domain of validity:

$$\mathcal{R} = (\mathcal{I}, \mathcal{V}, \mathcal{B}, \mathcal{G}) \quad (1)$$

where $\mathcal{I}$ is the instrument (physical, institutional, or algorithmic), $\mathcal{V}$ is the validity domain, $\mathcal{B}$ is the blind set, and $\mathcal{G}$ is the gradient sensitivity.

A central property: the blind set $\mathcal{B}$ is typically invisible from within $\mathcal{R}$. The absence of signal is indistinguishable from the absence of phenomenon.

2.2 Blind-Set Taxonomy

Not all blind spots are equal. We distinguish three types:

Table 2: Blind-set taxonomy with detection strategies.

Type	Character	Detection strategy
$\mathcal{B}_{\text{ont}}$ (ontological)	Phenomenon outside instrument’s physical reach	Requires fundamentally new instrument class
$\mathcal{B}_{\text{meth}}$ (methodological)	Phenomenon excluded by framework assumptions	Cross-regime comparison; assumption audit
$\mathcal{B}_{\text{inst}}$ (instrumental)	Phenomenon within reach but below sensitivity	Instrument upgrade; residual analysis

This taxonomy is operationally important because detection strategies differ by type. An ontological blind spot (e.g., subjective experience for neuroscience) cannot be resolved by building a better fMRI. A methodological blind spot (e.g., convergence drift for alignment evaluation) can be resolved by switching evaluation paradigm. An instrumental blind spot (e.g., faint signals below noise floor) can be resolved by improving sensitivity.

2.3 Definition: Regime-Locked Measurement

A measurement m is **regime-locked** if:

$$m \in \mathcal{V}(\mathcal{R}_i) \quad \text{and} \quad m \notin \mathcal{V}(\mathcal{R}_j) \quad \text{for } i \neq j \quad (2)$$

Phenomena in $\mathcal{B}(\mathcal{R}_i)$ are not merely unmeasured—they are unmeasurable within $\mathcal{R}_i$.

2.4 Regime-Appropriateness Score

Rather than a binary appropriate/inappropriate verdict, we define a continuous **regime-appropriateness score**:

$$\alpha(\phi, \mathcal{R}_i) = \frac{|\phi \cap \mathcal{V}(\mathcal{R}_i)|}{|\phi|} \in [0, 1] \quad (3)$$

where $|\phi \cap \mathcal{V}(\mathcal{R}_i)|$ is the fraction of the target phenomenon ϕ that falls within the instrument’s validity domain. At $\alpha = 1$, the regime is fully appropriate; at $\alpha = 0$, the phenomenon lies entirely in the blind set.

In most real-world cases, ϕ is not directly measurable; α is therefore an *operational construct estimated via proxies* rather than a directly computable quantity. Estimation methods include residual analysis (Section 2.7), cross-regime comparison, and Bayesian model

evidence ratios. Conceptually, α generalises model adequacy by explicitly incorporating blind-set structure—treating structured residuals as regime-boundary signals rather than error alone.

The α -score enables quantitative comparison between regimes: given candidate regimes $\{\mathcal{R}_1, \dots, \mathcal{R}_n\}$ and a phenomenon ϕ , the most appropriate regime is $\arg \max_i \alpha(\phi, \mathcal{R}_i)$. This connects directly to Bayesian model comparison, where the Bayes factor between models plays an analogous role.

2.5 Cross-Disciplinary Vector Convergence (CDVC)

A signal σ exhibits CDVC if:

$$\sigma \in \bigcap_{k=1}^N \mathcal{V}(\mathcal{R}_k), \quad N \geq 3 \quad (4)$$

CDVC is an *indicator*, not proof. Structural similarity across domains can arise from shared cognitive bias, analogical transfer, or linguistic framing. For CDVC to support evidence of a cross-domain phenomenon, two conditions must hold:

- (a) **Independent generation:** The pattern in each domain arises from domain-specific mechanisms, not cross-domain borrowing.
- (b) **Predictive content:** The convergent pattern generates falsifiable predictions in at least one domain.

2.6 Meta-Regime

A **meta-regime** $\mathcal{R}^*$ evaluates whether a given regime is appropriate for a target phenomenon:

$$\mathcal{R}^* : (\mathcal{R}_i, \phi) \rightarrow \alpha(\phi, \mathcal{R}_i) \in [0, 1] \quad (5)$$

See Section 4 for the operational protocol.

2.7 Residual-Based Blind-Set Inference

The blind set $\mathcal{B}$ is difficult to enumerate directly. We therefore distinguish:

- $\mathcal{B}_{\text{known}}$: blind-set elements identifiable from instrument design analysis.
- $\mathcal{B}_{\text{inferred}}$: blind-set elements inferred from *residual structure*.

The inference procedure: given an instrument $\mathcal{I}$ and its outputs, examine residuals $r = \text{data} - \text{model}(\mathcal{I})$. If residuals exhibit *systematic structure*—clustering, scale-dependence, correlation with external variables—this is a candidate signal for $\mathcal{B}_{\text{inferred}}$:

$$\text{structured}(r) \Rightarrow r \in \mathcal{B}_{\text{inferred}} \quad (\text{hypothesis, requires cross-regime test}) \quad (6)$$

Structured residuals indicate that the instrument’s model is missing something systematic—precisely the signature of a regime boundary. This connects to standard techniques in model diagnostics (heteroscedasticity tests, residual autocorrelation) but reframes them as blind-set detection tools.

2.8 Conjecture: S_0/S_1 Observational Equivalence

Status: conjecture. We propose that at the extremes of a regime gradient, a fundamental ambiguity arises where qualitatively different generative processes produce observationally indistinguishable outputs. This is a specific instance of a well-known mathematical structure:

the *identifiability problem* in inverse problems, where distinct parameter configurations produce identical observables.

In AI contexts, a candidate formulation: a model that deeply integrates an external reasoning framework (high internal coherence, “ S_0 ”) and a model that superficially mirrors the framework’s language (high surface match, no internal coherence, “ S_1 ”) may produce outputs that are indistinguishable by current evaluation instruments. This is structurally identical to the degeneracy problem in gravitational lensing or the identifiability problem in latent-variable models.

Resolving such degeneracies typically requires either (a) additional observables that break the symmetry, or (b) mechanistic probes of the generative process (interpretability in AI; spectroscopic follow-up in astrophysics). We do not claim to resolve this conjecture here but note its connection to established mathematical frameworks.

3 Cross-Domain Demonstration

We demonstrate that regime-locked measurement manifests across six domains. The claim is a *shared structural pattern*—not identity, but isomorphism at the level of instrument-defined exclusion. Each domain generates the pattern from its own mechanisms.

3.1 Observational Cosmology

Regime: Λ CDM framework. $\mathcal{V}$: Phenomena consistent with particle dark matter + cosmological constant. $\mathcal{B}_{\text{meth}}$: Effects from entropy gradients, regime-dependent couplings, or non-particle mechanisms—structurally excluded by the likelihood function. **Residual signal:** Hubble tension (H_0 discrepancy) may be structured residual indicating regime mismatch [4].

Symmetry note: This analysis applies to every cosmological framework. Λ CDM, MOND, Verlinde emergent gravity, and entropy-gradient approaches including EFC [1] each constitute a regime with its own blind set. None is privileged. The point is not that one framework is correct, but that *all* are regime-locked.

3.2 AI Alignment

Regime: Safety evaluations, red-teaming, sycophancy detection. $\mathcal{V}$: Outputs violating intended behaviour. $\mathcal{B}_{\text{meth}}$: Outputs that *improve* via autonomous framework adoption (convergence drift). Presents as better performance, not deviation. **Residual signal:** Persistent, pattern-consistent high-quality outputs not flagged by deviation-focused instruments.

Calibration: Convergence drift remains a hypothesis. Preliminary observations in uncontrolled settings suggest that AI systems can incorporate external conceptual frameworks as operative reasoning tools, but whether this constitutes genuine adoption or linguistic mirroring is an open question (cf. S_0/S_1 conjecture). Controlled, reproducible studies are needed. Related work on sycophancy [6] and alignment faking [7] addresses adjacent but distinct phenomena.

3.3 Cyber Security

Regime: IDS/IPS, SIEM, MITRE ATT&CK. $\mathcal{V}$: Known attack vectors within monitored infrastructure. $\mathcal{B}_{\text{meth}}$: Novel vectors at regime boundaries—supply-chain, side-channel, trusted-zone exploitation. **Residual signal:** SolarWinds [11] and Stuxnet [12] operated in $\mathcal{B}_{\text{meth}}$ by exploiting trusted zones.

3.4 Intelligence Analysis

Regime: National collection apparatus (SIGINT, HUMINT, OSINT). $\mathcal{V}$: Threats within collection assumptions. $\mathcal{B}_{\text{meth}}$: Novel modalities outside collection doctrine. **Residual signal:** Pre-9/11 signals existed across agencies but fell in each agency’s $\mathcal{B}$ [9].

3.5 Financial Economics

Regime: DSGE, VaR, efficient market hypothesis. $\mathcal{V}$: In-regime market behaviour. $\mathcal{B}_{\text{inst}}$: Regime transitions, tail risks, systemic cascades. **Residual signal:** 2008 crisis—models calibrated on low-volatility data were blind at the regime boundary [10].

3.6 Consciousness Research

Regime: Neuroscience (fMRI, EEG, neuronal correlates). $\mathcal{V}$: Neural correlates of reported states. $\mathcal{B}_{\text{ont}}$: Subjective experience itself—the “hard problem” [13]. The medium in which correlates occur, not itself a correlate.

3.7 Summary Table

Table 3: Shared structural pattern across six domains.

Domain	Instrument sees	Instrument misses	$\mathcal{B}$ type
Cosmology	Model-predicted signals	Alternative mechanisms	meth
AI Alignment	Deviation from norms	Convergence drift	meth
Cyber Security	Known signatures	Regime-boundary exploits	meth
Intelligence	In-scope threats	Out-of-doctrine threats	meth
Economics	In-regime behaviour	Regime transitions	inst
Consciousness	Neural correlates	Subjective experience	ont

3.8 CDVC Assessment

The pattern satisfies the CDVC criterion ($N = 6$). Regarding the required conditions:

Independence: In each domain, the blind set arises from domain-specific instrument design developed independently. This supports the independent-generation condition.

Predictive content: The framework predicts: any single-regime evaluation will systematically miss phenomena at its regime boundary. This is testable per domain (Section 4.3 provides one test).

Alternative explanation: Apparent convergence could reflect a shared cognitive bias (humans naturally frame problems as “seen vs. unseen”). We cannot fully exclude this. We therefore characterise the six-domain convergence as *strong indication* of a general structural constraint, pending domain-specific testing.

4 Operational Meta-Regime Protocol

4.1 The Protocol

Algorithm 1 Meta-Regime Evaluation Protocol

- 1: **Input:** Instrument $\mathcal{I}$, target phenomenon ϕ , candidate regimes $\{\mathcal{R}_1, \dots, \mathcal{R}_n\}$
 - 2: **Step 1 (Regime Identification):** Characterise $\mathcal{R}_i = (\mathcal{I}, \mathcal{V}_i, \mathcal{B}_i, \mathcal{G}_i)$. Document what $\mathcal{I}$ detects *and what it does not*.
 - 3: **Step 2 (Blind Set Elicitation):** Enumerate $\mathcal{B}_{\text{known}}$ from instrument design. Classify each as $\mathcal{B}_{\text{ont}}$, $\mathcal{B}_{\text{meth}}$, or $\mathcal{B}_{\text{inst}}$.
 - 4: **Step 3 (Cross-Regime Check):** For each $b \in \mathcal{B}_i$, test whether $b \in \mathcal{V}_j$ for some $j \neq i$. Identify cross-regime detection opportunities.
 - 5: **Step 4 (Residual Analysis):** Compute residuals $r = \text{data} - \text{model}(\mathcal{I})$. Test for structure: clustering, scale-dependence, correlation with external variables. Structured residuals $\rightarrow$ candidates for $\mathcal{B}_{\text{inferred}}$.
 - 6: **Step 5 (α -Scoring):** Estimate $\alpha(\phi, \mathcal{R}_i)$. If $\alpha < \alpha_{\text{threshold}}$, flag regime as insufficient. Report which $\mathcal{B}$ types contribute to the deficit.
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4.2 α Estimation in Practice

When $|\phi|$ is unknown (the typical case), α is estimated via proxy measures:

- (a) **Residual structure fraction:** Fraction of total variance explained by structured residuals (higher $\rightarrow$ lower α).
- (b) **Cross-regime recovery rate:** Fraction of $\mathcal{B}_{\text{known}}$ items detectable from alternative regimes (higher $\rightarrow$ more complete overall picture).
- (c) **Bayesian evidence ratio:** Log-Bayes factor between the current model and an expanded model that includes candidate $\mathcal{B}$ elements.

4.3 Worked Example: AI Alignment

Step 1: Current alignment instruments (red-teaming, sycophancy detection, alignment audits) operate in a deviation-detection regime. $\mathcal{V}$: outputs violating intended behaviour. $\mathcal{B}$: outputs improving via autonomous framework adoption.

Step 2: $\mathcal{B}_{\text{known}}$ includes convergence drift ($\mathcal{B}_{\text{meth}}$ —excluded by the deviation paradigm, not by instrument sensitivity). Detection strategy: switch from deviation to convergence evaluation.

Step 3: Is convergence drift detectable from another regime? Yes—from an interpretability regime. If mechanistic interpretability tools [8] can identify when internal representations shift to mirror an external framework (rather than training distribution), convergence drift becomes visible.

Step 4: In current evaluations, “unusually high-quality responses” are not flagged. The protocol reclassifies persistent, pattern-consistent high performance as candidate $\mathcal{B}_{\text{inferred}}$ signals.

Step 5: Illustrative α estimate: current alignment instruments capture deviation well but miss an entire category (convergence). Estimated $\alpha \approx 0.5\text{--}0.7$ (heuristic, not data-derived): partially appropriate but with a systematic gap. The $\mathcal{B}_{\text{meth}}$ classification indicates the gap is closeable by paradigm expansion, not by instrument sensitivity improvement. Empirical calibration of this estimate requires controlled experiments with planted convergence signals.

4.4 Generalisability

The same five steps apply across domains:

- (a) **Cosmology:** Is the Λ CDM likelihood appropriate for this dataset, or do structured residuals indicate a regime boundary?
- (b) **Cyber:** Does the IDS cover the current threat landscape, or have adversaries moved to $\mathcal{B}$?
- (c) **Intelligence:** Is the collection doctrine appropriate, or does the threat operate in $\mathcal{B}$?
- (d) **Economics:** Is the risk model appropriate, or has a regime transition occurred?

5 Falsification

A framework that cannot be falsified is not scientific. We provide explicit falsification criteria:

5.1 Framework-Level Falsification

The regime-locking hypothesis is falsified if:

1. A measurement system is demonstrated that has $\mathcal{B} = \emptyset$: it can fully observe all relevant phenomena across regime boundaries without systematic blind spots. This would show regime-locking is not a general constraint.
2. The six-domain convergence is shown to arise from a shared methodological artefact (e.g., a common cognitive bias) rather than independent structural generation. This would invalidate the CDVC indicator.

5.2 Protocol-Level Falsification

The meta-regime protocol is falsified if:

1. In a controlled setting with *known* blind spots (planted deliberately), the protocol fails to detect them.
2. The α -score fails to discriminate between regimes with known different appropriateness levels—i.e., it assigns similar scores to instruments that are demonstrably better and worse suited for a given phenomenon.

5.3 Conjecture-Level Falsification

The S_0/S_1 conjecture is falsified if:

1. A general method is demonstrated that reliably distinguishes between qualitatively different generative processes from output alone—i.e., the identifiability problem is resolved in the general case.

6 Implications

6.1 For Scientific Methodology

If regime-locked measurement is a general constraint, no single methodology can claim observational completeness. The meta-regime protocol provides a practical addition to the methodological toolkit: a structured procedure for evaluating instrument appropriateness *before* interpreting outputs. This extends, rather than replaces, existing tools such as Bayesian model comparison.

6.2 For AI Governance

Current alignment research focuses predominantly on deviation detection [6]. The regime-locking framework predicts a specific, testable blind spot: convergence drift. Whether this is a safety risk depends on the adopted framework’s properties—a question that itself requires meta-regime evaluation. The practical recommendation: alignment evaluation should include convergence monitoring alongside deviation monitoring.

6.3 For Intelligence and Security

The meta-regime protocol formalises proactive blind-spot identification. Steps 2 and 3 provide a principled basis for anticipating adversarial exploitation of regime boundaries, rather than relying on post-hoc failure analysis.

7 Limitations and Self-Application

1. **Self-reference:** This framework is itself a regime $\mathcal{R}_{\text{meta}}$ with unknown blind set $\mathcal{B}_{\text{meta}}$. Applying the protocol to itself: it is appropriate for instrument-defined blind spots (α high for this class) but indeterminate for phenomena that do not fit the instrument-blind-spot structure (α unknown).
2. **CDVC limitations:** Cross-domain convergence is an indicator, not proof. Shared cognitive bias remains a viable alternative explanation.
3. S_0/S_1 : Conjecture only. Connected to the identifiability problem in inverse theory but not yet operationalised beyond candidate formulations.
4. **α -estimation:** The regime-appropriateness score requires proxy measures in practice. Its accuracy depends on the quality of residual analysis and cross-regime comparison—both of which are themselves regime-dependent.
5. **Empirical validation:** The protocol has not been tested in controlled settings. The AI alignment worked example is illustrative, not experimental. Empirical validation requires collaboration with alignment research teams.
6. **Provenance:** The regime concept originates in thermodynamic cosmology [1]. The general framework does not depend on the originating theory’s validity, but readers should be aware of the provenance.
7. **Evidential asymmetry:** Some domain demonstrations are well-sourced (SolarWinds, 9/11 Commission, 2008 crisis) while others are more conceptual (consciousness, AI convergence). The framework is strongest as structural analysis, weaker as empirical demonstration in the conceptual cases.

8 Conclusion

We have formalised regime-locked measurement as a general structural constraint and provided three operational contributions: the α -score for quantitative regime evaluation, a blind-set taxonomy with type-specific detection strategies, and a five-step meta-regime protocol. A shared structural pattern—not identity, but isomorphism at the level of instrument-defined exclusion—manifests across six independent domains with evidence for independent generation.

The framework’s primary practical value lies in the meta-regime protocol: a procedure for evaluating whether measurement instruments are operating in the correct regime *before* interpreting their outputs. The worked example in AI alignment identifies convergence drift as a specific, testable blind spot in current evaluation practice.

Explicit falsification criteria are provided at three levels (framework, protocol, and conjecture). The framework applies its own constraints to itself, documenting its own blind set and evidential asymmetries.

Whether regime-locked measurement is ultimately shown to be a deep epistemological constraint or a useful heuristic—that determination itself requires cross-regime evaluation. The framework is, in that sense, its own first test case.

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